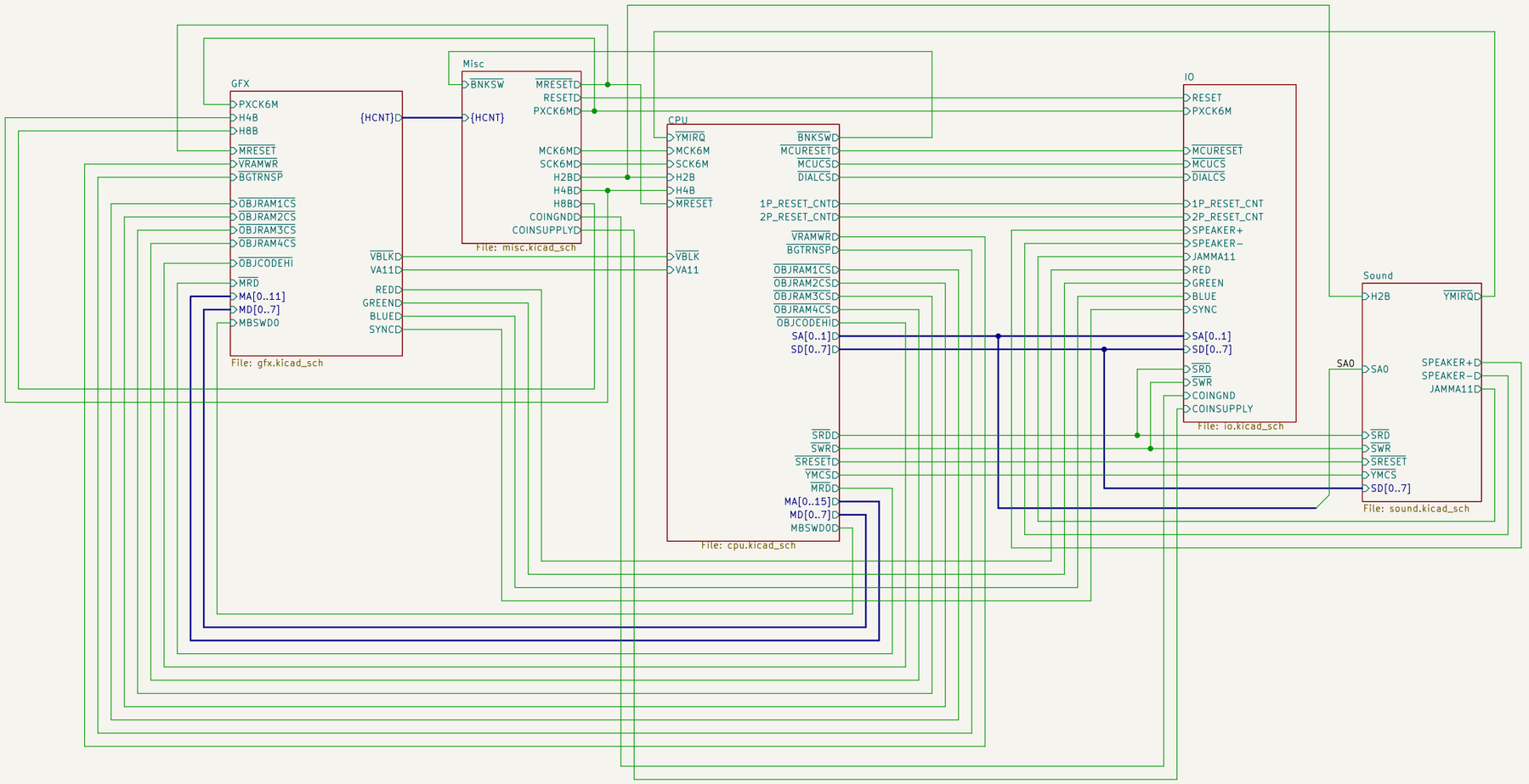


Seta number: P0-022-A

Ulf Skutnabba, twitter: @skutis77

Sheet: /

File: arknoid2.kicad_sch

Title: Arkanoid: Revenge of Doh

Size: A3	Date: 2023-02-22	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/6

E9:B06-10

-VRAMWR = MA15 & MA14 & MA13 & MA12 & ~MA11 & ~MA10 & ~MMREQ & MRFSH
-BGTRNSP = MA15 & MA14 & MA13 & MA12 & ~MA11 & MA10 & ~MA9 & ~MMREQ & MRFSH & ~MWR
-BNKSW = MA15 & MA14 & MA13 & MA12 & ~MA11 & MA10 & MA9 & ~MMREQ & MRFSH & ~MWR
-OBJCODELO = MA15 & MA14 & ~MA13 & ~MA12 & ~MMREQ & MRFSH
-OBJCODEHI = MA15 & MA14 & ~MA13 & MA12 & ~MMREQ & MRFSH
-MSHRAMRQ = MA15 & MA14 & MA13 & ~MA12 & ~MMREQ & MRFSH
-OBJORVRAMEN = MA15 & MA14 & MA13 & MA12 & ~MA11 & ~MMREQ & MRFSH & (~MA9 & ~MWR + ~MA10) + MA15 & MA14 & ~MA13 & ~MMREQ & MRFSH

D6:B06-11

-MROMCS = ~MMREQ & MRFSH & (~MA15 & ~MA14 & MR2A14 + ~MA14 & ~MR2A14 + ~MA15 & MA14)
-MROM2CS = MA15 & ~MA14 & ~MMREQ & MRFSH & MR2A14
MRA14 = MA15 & MBSWD0 + ~MA15 & MA14
-OBJRAM4CS = VA11 & (OBJCODELO & H4B + ~OBJCODELO)
-OBJRAM3CS = ~VA11 & (OBJCODELO & H4B + & ~OBJCODELO)
-OBJRAM2CS = VA11 & (H4B + ~OBJCODEHI)
-OBJRAM1CS = ~VA11 & (H4B + ~OBJCODEHI)

C3:B06-12
i7 and i8 have been set to logical one in the equations.

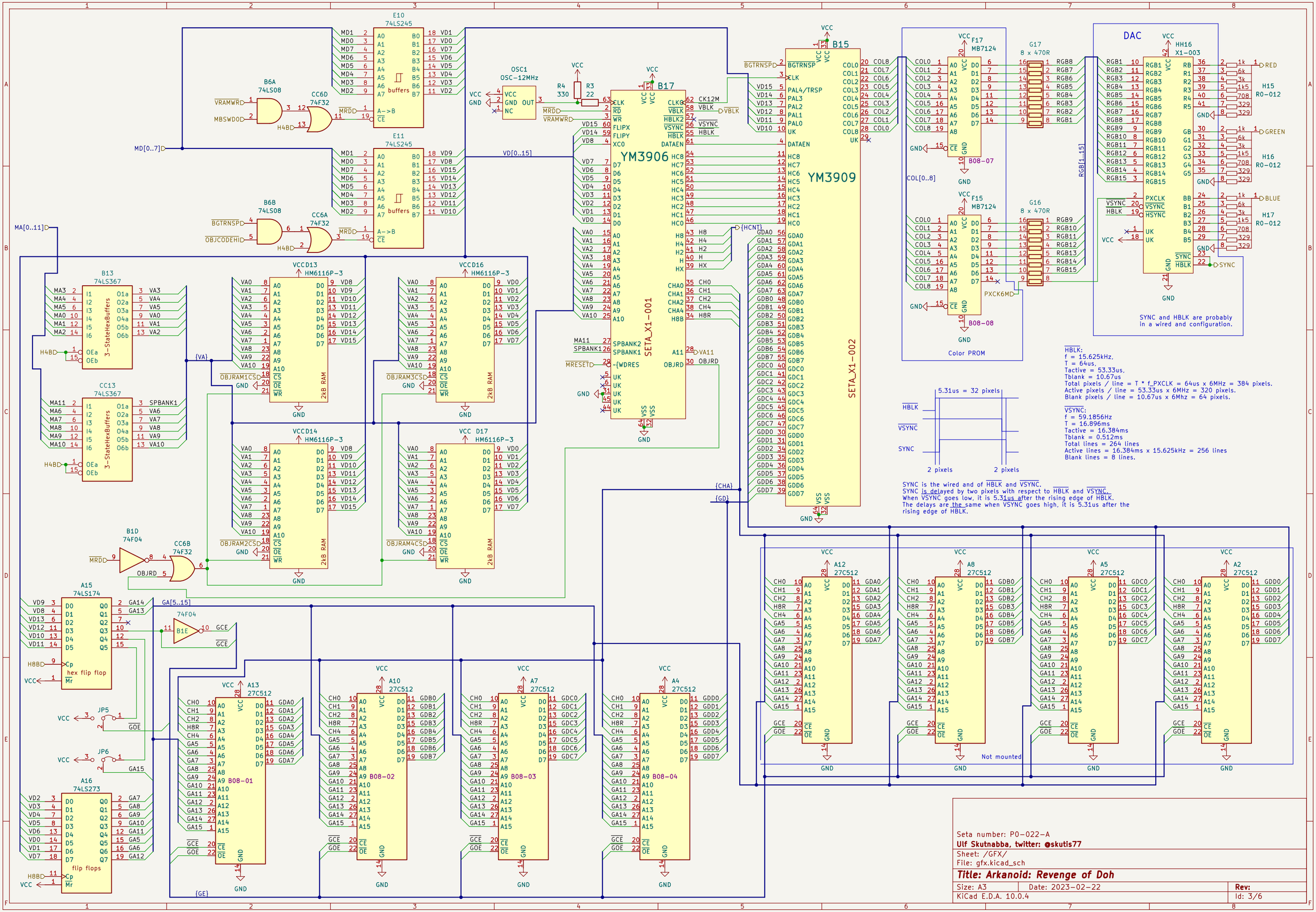
SWAIT = YMCS2 & ~YMCS & (~SSHRAMEN + SSHRAMRQ)
MWAIT = SSHRAMEN & (OBJORVRAMEN + H2H4)
SSHRAMEN = SSHRAMRQ + ~MSHRAMRQ & SSHRAMEN
MSHRAMEN = MSHRAMRQ + ~SSHRAMRQ & MSHRAMEN
SHRAMCS = MSHRAMRQ & SSHRAMRQ
SHRAMWR = MWR & SSHRAMEN + MSHRAMRQ & SSHRAMRQ + SWR & MSHRAMEN
SHRAMENB = SSHRAMEN

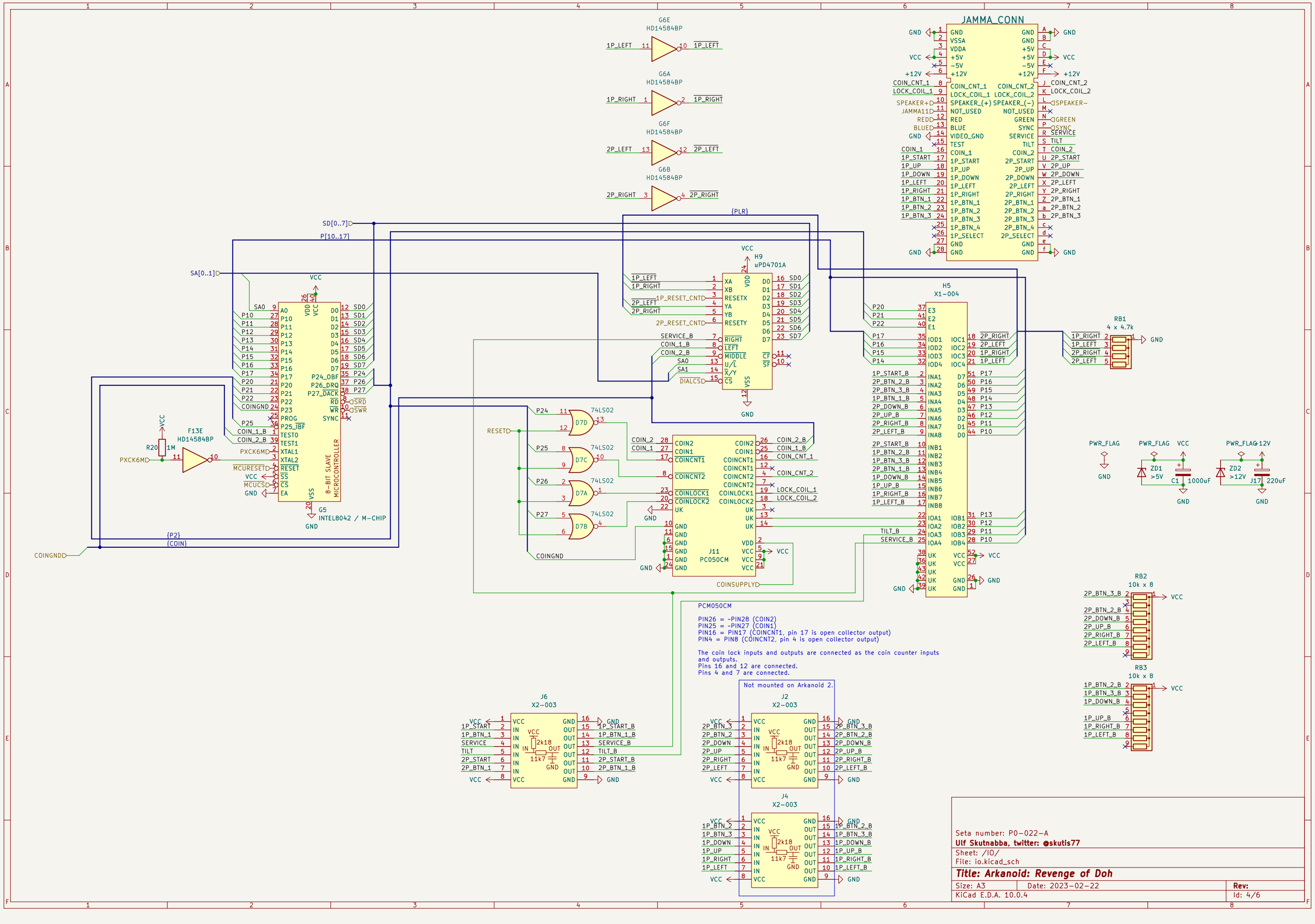
C2:B06-13

-SROMCS = ~SMREQ & SRFSH & (~SA14 & ~SA13 + ~SA15 & SA14 + ~SA15 & ~SA14)
-SSHRAMRQ = ~SMREQ & SRFSH & (SA15 & SA14 & SA13 & ~SA12 + SA15 & SA14 & ~SA13 & SA12)
-SBNKSW = SA15 & ~SA14 & SA13 & ~SA12 & ~SWR & ~SMREQ & SRFSH
-YMCS = SA15 & ~SA14 & SA13 & SA12 & ~SMREQ & SRFSH
-MCUCS = SA15 & SA14 & ~SA13 & ~SA12 & ~SMREQ & SRFSH
-DIALCS = SA15 & SA14 & SA13 & SA12 & ~SMREQ & SRFSH
SRA13 = SA15 & ~SBANK0 + ~SA15 & SA13
SRA14 = SA15 & ~SBANK1 + ~SA15 & SA14

Not mounted

Seta number: P0-022-A			
Ulf Skutnabba, twitter: @skutis77			
Sheet: /CPU/			
File: cpu.kicad_sch			
Title: Arkanoid: Revenge of Doh			
Size: A3	Date: 2023-02-22	**Rev:**	
KiCad E.D.A. 10.0.4		Id: 2/6	


E10
74LS245
buffers
A->B
B6A
74LS08
CC6D
74F32
VRAMWRD
MBSWD0
H4BD
MRDD
MD[0..7]
E11
B6B
BGTRNSP
OBJCODEHI
CC6A
OSC1
OSC-12MHz
R4
330
R3
22
VD[0..15]
B17
YM3906
SETA_X1-001
CLK
RD
WR
FLIPX
FLIPY
XCO
CK12M
VBLK
HBLK2
VSYNC
HBLK
DATAEN
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
H8
H4
H2
H
HX
CHA0
CHA1
CHA2
CHA4
H8B
A11
OBJRD
SPBANK2
SPBANK1
~{WDRES
MRESET
B15
YM3909
SETA_X1-002
PAL4/TRSP
PAL3
PAL2
PAL1
PAL0
UK
GDA0
GDA1
GDA2
GDA3
GDA4
GDA5
GDA6
GDA7
GDB0
GDB1
GDB2
GDB3
GDB4
GDB5
GDB6
GDB7
GDC0
GDC1
GDC2
GDC3
GDC4
GDC5
GDC6
GDC7
GDD0
GDD1
GDD2
GDD3
GDD4
GDD5
GDD6
GDD7
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
{HCNT}
{CHA}
{GD}
F17
MB7124
B08-07
F15
B08-08
Color PROM
COL[0..8]
G17
8 x 470R
G16
RGB[1..15]
RGB8
RGB7
RGB6
RGB5
RGB4
RGB3
RGB2
RGB1
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
PXCK6M
DAC
HH16
X1-003
RED
GREEN
BLUE
H15
R0-012
H16
R0-012
H17
R0-012
PXCLK
VSYNC
HBLK
SYNC
SYNC and HBLK are probably
in a wired and configuration.
HBLK:
f = 15.625kHz,
T = 64us,
Tactive = 53.33us,
Tblank = 10.67us
Total pixels / line = T * f_PXCLK = 64us x 6MHz = 384 pixels.
Active pixels / line = 53.33us x 6MHz = 320 pixels.
Blank pixels / line = 10.67us x 6Mhz = 64 pixels.
VSYNC:
f = 59.1856Hz
T = 16.896ms
Tactive = 16.384ms
Tblank = 0.512ms
Total lines = 264 lines
Active lines = 16.384ms x 15.625kHz = 256 lines
Blank lines = 8 lines.
5.31us = 32 pixels
2 pixels
2 pixels
SYNC is the wired and of HBLK and VSYNC.
SYNC is delayed by two pixels with respect to HBLK and VSYNC.
When VSYNC goes low, it is 5.31us after the rising edge of HBLK.
The delays are the same when VSYNC goes high, it is 5.31us after the
rising edge of HBLK.
MA[0..11]
B13
74LS367
3-StateHexBuffers
CC13
{VA}
VCCD13
VCCD16
VCCD14
VCC D17
HM6116P-3
2kB RAM
OBJRAM1CS
OBJRAM3CS
OBJRAM2CS
OBJRAM4CS
B1D
74F04
CC6B
74F32
A15
74LS174
hex flip flop
H8BD
A16
74LS273
flip flops
GA[5..15]
B1E
GCE
JP5
JP6
GOE
GA15
{GE}
A13
27C512
B08-01
A10
B08-02
A7
B08-03
A4
B08-04
A12
A8
A5
A2
Not mounted
Seta number: P0-022-A
Ulf Skutnabba, twitter: @skutis77
Sheet: /GFX/
File: gfx.kicad_sch
Title: Arkanoid: Revenge of Doh
Size: A3
Date: 2023-02-22
KiCad E.D.A. 10.0.4
Rev:
Id: 3/6


JAMMA_CONN
GND
VCC
+12V
COIN_CNT_1
LOCK_COIL_1
SPEAKER+
JAMMA11D
REDD
BLUED
COIN_1
1P_START
1P_UP
1P_DOWN
1P_LEFT
1P_RIGHT
1P_BTN_1
1P_BTN_2
1P_BTN_3
1P_BTN_4
1P_SELECT
VSSA
VDDA
+5V
−5V
NOT_USED
RED
BLUE
VIDEO_GND
TEST
SPEAKER_(+)
SPEAKER_(−)
GREEN
SYNC
SERVICE
TILT
COIN_2
2P_START
2P_UP
2P_DOWN
2P_LEFT
2P_RIGHT
2P_BTN_1
2P_BTN_2
2P_BTN_3
2P_BTN_4
2P_SELECT
SPEAKER−
G6E
HD14584BP
1P_LEFT
G6A
1P_RIGHT
G6F
2P_LEFT
G6B
2P_RIGHT
SD[0..7]
P[10..17]
SA[0..1]
{PLR}
H9
uPD4701A
1P_RESET_CNT
2P_RESET_CNT
SERVICE_B
COIN_1_B
COIN_2_B
SA0
SA1
DIALCS
RIGHT
LEFT
MIDDLE
U/D
X/Y
CS
CF
SF
H5
X1−004
IOD1
IOD2
IOD3
IOD4
IOC1
IOC2
IOC3
IOC4
INA1
INA2
INA3
INA4
INA5
INA6
INA7
INA8
INB1
INB2
INB3
INB4
INB5
INB6
INB7
INB8
IOA1
IOA2
IOA3
IOA4
IOB1
IOB2
IOB3
IOB4
RB1
4 x 4.7k
F13E
HD14584BP
R20
1M
PXCK6MD
MCURESETD
MCUCSD
P25_IBF
TESTO
TEST1
XTAL1
XTAL2
RESET
SS
CS
EA
8−BIT SLAVE MICROCONTROLLER
G5
INTEL8042 / M−CHIP
{P2}
{COIN}
COINGND
RESETD
74LS02
D7D
D7C
D7A
D7B
COINCNT1
COINCNT2
COINLOCK1
COINLOCK2
UK
J11
PC050CM
COIN_CNT_1
COIN_CNT_2
LOCK_COIL_1
LOCK_COIL_2
COINSUPPLY
PCM050CM
PIN26 = -PIN28 (COIN2)
PIN25 = -PIN27 (COIN1)
PIN16 = PIN17 (COINCNT1, pin 17 is open collector output)
PIN4 = PIN8 (COINCNT2, pin 4 is open collector output)
The coin lock inputs and outputs are connected as the coin counter inputs and outputs.
Pins 16 and 12 are connected.
Pins 4 and 7 are connected.
Not mounted on Arkanoid 2.
J6
X2−003
J2
J4
2k18
11k7
PWR_FLAG
ZD1
>5V
C1
1000uF
ZD2
>12V
J17
220uF
RB2
10k x 8
RB3
Seta number: P0−022−A
Ulf Skutnabba, twitter: @skutis77
Sheet: /IO/
File: io.kicad_sch
Title: Arkanoid: Revenge of Doh
Size: A3
Date: 2023−02−22
KiCad E.D.A. 10.0.4
Rev:
Id: 4/6

SCPU controls databus if it is floating or not.

SD[0..7]

SD0 SD1 SD2 SD3 SD4 SD5 SD6 SD7

SA0D

SRDD SWRD YMCSD SRESETD

H2BD

YMIRQ

VCC

G9
YM2203

SWA
SW_DIP_x08

SWB
SW_DIP_x08

GND

H11
YM3014

H12B
H12A
RC4558

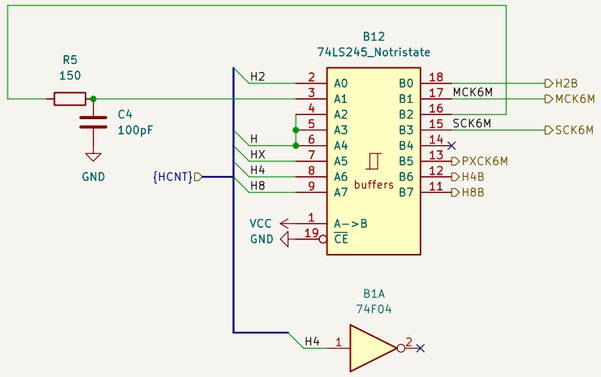
R7 1k
C9 10uF
C10 100nF
R9 8.2k
R21 1k
C12 10nF
VR1 890R
C13 1uF
C11 4.7uF
R8 6.2k
JAMMA11

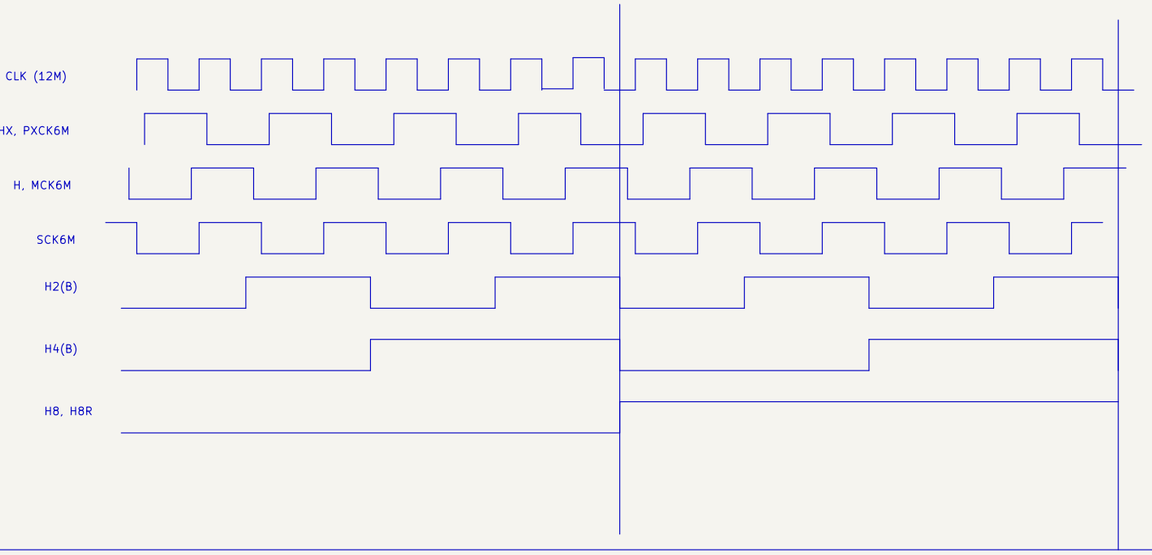
J16
HA1388
18 W BTL Audio Power Amplifier
Amp1
Amp2
ASO_Protect
Surge Protect
TSD

C18 47uF
C17 47uF
C15 47uF
C16 47uF
C19 47uF
C14 47uF
R10 2.2
R11 2.2
C20 100nF
C21 100nF
SPEAKER+
SPEAKER−

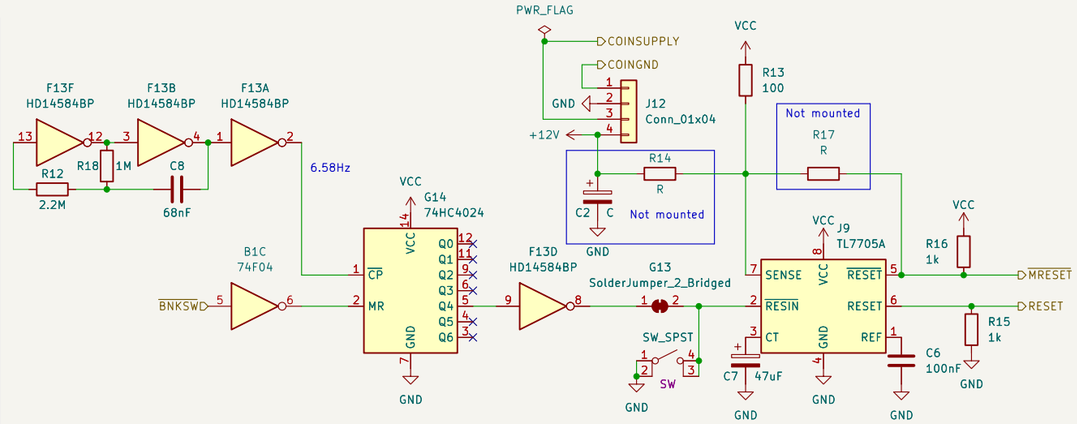
Seta number: P0-022-A		
Ulf Skutnabba, twitter: @skutis77		
Sheet: /Sound/		
File: sound.kicad_sch		
Title: Arkanoid: Revenge of Doh		
Size: A3	Date: 2023-02-22	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/6

Clocks





During normal operation pin 1 is shorted to pin 2 and pin 3 is shorted to pin 4.



Seta number: PO-022-A
Ulf Skutnabba, twitter: @skutis77

Sheet: /Misc/
File: misc.kicad_sch

Title: Arkanoid: Revenge of Doh

Size: A3	Date: 2023-02-22	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/6